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Senior High School

Ratio and Proportion

**Learner's Module in
Business Mathematics**
Quarter 1 • Module 3 • Week 3



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NAME: _____ GRADE AND SECTION _____
TEACHER: _____ SCORE _____

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What I Know



If you answer all the test items correctly in this pretest, then you may skip studying this learning material and proceed to the next learning module.

Multiple Choice: Read and understand each item carefully. Choose the letter of the BEST answer. Write your answers on your answer sheet.

1. Mario had 28 kg of strawberries for sale. He sold 7 kg of the strawberries to a friend. What was the ratio of the mass of strawberries sold to the mass of strawberries left?
A. 1:3 B. 1:4 C. 7:21 D. 7:28
2. Express the ratio 162:126:90 in simplest form.
A. 81:63:45 C. 9:7:5
B. 27:21:15 D. 3:7:5
3. Which of the following ratios are equivalent?
A. 300 miles in 6 hours and 200 miles in 5 hours.
B. 8 km in 21 minutes and 12 km in 31.5 minutes.
C. 5 adults to accompany 25 children and 3 adults to accompany 12 children.
D. 5 glasses of iced tea for 180 grams of powdered mix and 2 glasses of iced tea for 36 grams of powdered mix.
4. What value of x will make the fractions $\frac{x}{21}$ and $\frac{171}{399}$ equivalent?
A. 3 B. 9 C. 18 D. 57
5. How many meters of ribbon will be needed for a dozen gifts, if a gift box needs $\frac{3}{8}$ m of ribbon?
A. 4.5 m B. 9 m C. 18 m D. 32 m
6. Which of the following statements is true about the kinds of proportion?
A. A number is directly proportionate to another when as one value increases, the other decreases.
B. A number is indirectly proportionate to another when as one value increases, so does the other.
C. Partitive proportion involves identifying parts of a whole based on a given ratio of these parts.
D. Direct proportion has two values that may vary so the ratio between them also varies.
7. A type of proportion in which the quantities are inversely related with each other.
A. Direct Proportion C. Mere Proportion
B. Indirect Proportion D. Partitive Proportion

8. Which of the given quantities show direct relationship?
- | | |
|-------------------------------|------------------------|
| A. Time and Speed | C. Volume and Pressure |
| B. Number of workers and days | D. Quantity and Cost |
9. What kind of proportion is exemplified in the situation: "Three businessmen share profits and losses at a ratio of 2:3:5."?
- | | |
|-------------|--------------|
| A. Direct | C. Mere |
| B. Indirect | D. Partitive |
10. Which of the following real-life examples is true about proportion?
- A. Quantity and cost are indirectly related.
 B. Time and speed are directly related.
 C. Number of workers and days is under partitive proportion.
 D. Budgeting a monthly salary is an example of partitive proportion.
11. Kobe resells face masks and face shields in this time of pandemic in the ratio 8:3. If he sold 121 facemasks and face shields for the past week, how many face shields did he sell?
- | | | | |
|-------|-------|-------|-------|
| A. 22 | B. 33 | C. 66 | D. 88 |
|-------|-------|-------|-------|
12. The quality control team of a toy company accounted that there are 5 defective toy models for every 600 models. If the company makes 66 000 toy models annually, how many of them are expected to be defective?
- | | | | |
|--------|--------|--------|----------|
| A. 110 | B. 120 | C. 550 | D. 3 000 |
|--------|--------|--------|----------|
13. Sam bought 15 cones of ice cream with 3 scoops of different flavors in each cone. How many scoops of ice cream will there be if there are 120 cones?
- | | | | |
|-------|-------|--------|--------|
| A. 24 | B. 40 | C. 360 | D. 600 |
|-------|-------|--------|--------|
14. Ciara and her friends started to save money to be donated to the front liners in their province. In two weeks, they were able to save ₱ 4 000. With this rate of saving money, how many weeks will they need to be able to raise ₱ 52 000?
- | | |
|-------------|-------------|
| A. 26 weeks | C. 15 weeks |
| B. 24 weeks | D. 13 weeks |
15. Justine and Ethan agreed to go into a partnership. The former invested ₱ 60 000 while the latter invested ₱ 120 000. What part of the total capital is that of Ethan?
- | | | | |
|------------------|------------------|------------------|------------------|
| A. $\frac{2}{3}$ | B. $\frac{1}{2}$ | C. $\frac{1}{4}$ | D. $\frac{1}{3}$ |
|------------------|------------------|------------------|------------------|



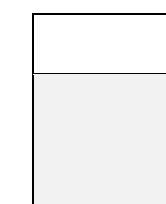
What's In



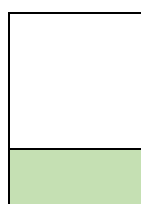
It cannot be denied that in our everyday life, numbers are often used to compare quantities. For example, a mixture of calamansi extract and water as our drink is proven to boost our immune system especially in the midst of the COVID-19 pandemic. Shown below are ratios of the mixture, however, a change in portioning of the ingredients affects the taste of the drink.

Calamansi juice: **7** parts of water, **3** parts of calamansi extract.

The **ratio** of the number of parts of water to the number of parts of calamansi extract used is **7 to 3**. Written as **7:3**. Let's take a look this simple experiment.



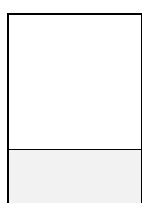
7 parts of water



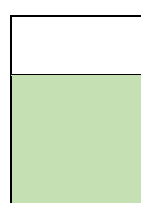
3 parts of calamansi extract



It's just nice!



3 parts of water



7 parts of calamansi extract



It's too sour!

Definitely, 7 parts of water to 3 parts of calamansi extract is different from 3 parts of water to 7 parts of calamansi extract. Thus, **3:7 is not equal to 7:3**.

The above situation and its relation to proportion will be dealt with extensively in this module. Are we ready for more? Let's get going!



What's New



Activity #1: Help Me Write Ratios

The table below shows the masses of fruits sold in one fruit stall.

Vegetable	Avocado	Banana	Cucumber	Durian	Papaya
Mass	28 kg	32 kg	49 kg	14 kg	40 kg

Direction: Complete the table below by writing ratios on the masses of the fruits in any combination and simplify each written ratio. Use a separate sheet for your answers. An example is given for you to follow.

	RATIO	SIMPLEST FORM
Example: The ratio of mass of avocado to mass of durian.	28 : 14	2 : 1
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		



LESSON 1: RATIO

Lesson 1.1: A Glimpse of Ratio

Ratio is a comparison of two or more positive quantities. It shows the relative sizes of two or more values or the relation between two or more magnitudes of the same kind, and ratio has no unit.

When we say, for example, that there are twice as many females as males contracted by COVID-19, we mean that the ratio of females to males contracted by the virus is 2:1 in which 2 represents females and 1 represents males.

Illustrative examples:

1. Andrei and Roey are presidential candidates in their school Supreme Student Government. Andrei received 250 votes while Roey had 300 votes. What is the ratio of Andrei's vote against Roey?

Solution:

$$\text{Andrei} : \text{Roey} = 250 : 300 = 25 : 30 = 5 : 6$$

Thus, the ratio of Andrei's votes to Roey's votes is 5:6.

Checking:

$$\text{Andrei's votes} + \text{Roey's votes} = 550$$

$$\frac{5}{11} (550) + \frac{6}{11} (550) = 550 \quad *** (5 + 6 = 11, \text{the denominator})$$

$$250 + 300 = 550$$

$$550 = 550$$

2. Leo, Mel, and Neo are business partners. Their capital balances are ₱ 10 000.00, ₱ 20 000.00, and ₱ 30 000.00 respectively. What is their capital ratio?

Solution:

$$\text{Leo} : \text{Mel} : \text{Neo} = 10\,000 : 20\,000 : 30\,000 = 1 : 2 : 3$$

Thus, the capital ratio of Leo, Mel and Neo is 1:2:3.

3. A cash inheritance is divided among 4 children in such a way that Art got $\frac{1}{5}$, Ben got $\frac{3}{10}$, Cass got $\frac{1}{10}$, and Drew got $\frac{2}{5}$. Find the ratio in which the four children divided the cash inheritance.

Solution:

There is a need to change the dissimilar fractions to similar ones.

$$\text{Art} = \frac{1}{5} \Rightarrow \frac{2}{10}$$

$$\text{Ben} = \frac{3}{10}$$

$$\text{Cass} = \frac{1}{10}$$

$$\text{Drew} = \frac{2}{5} \Rightarrow \frac{4}{10}$$

Therefore, the ratio in which the four children divided the cash inheritance is 2 : 3 : 1 : 4.

***The numerators of the similar fractions were taken to form the ratio.

4. Liquidity and Solvency Ratios

Liquidity ratio gauges if a company can pay its liabilities for the current year, while solvency ratio is concerned with meeting the long-term obligations of the company. **Current ratio** measures the liquidity of a company, **debt ratio** on the other hand measures the solvency of a company. Below are the formulae you can use in analyzing the liquidity and solvency of a company.

$$\text{a. Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}.$$

$$\text{b. Debt Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

Illustrative example:

The following amounts were taken from Delta Company's financial reports:

Current Assets	₱ 60 000.00
Total Assets	₱ 700 000.00
Current Liabilities	₱ 20 000.00
Total Liabilities	₱ 350 000.00

- What is the current ratio? What does this ratio mean (provide a short interpretation of the computed ratio)?
- What is the debt ratio? What does this ratio mean (provide a short interpretation of the computed ratio)?

Solution:

$$\begin{aligned} \text{a. Current Ratio} &= \frac{\text{Current Assets}}{\text{Current Liabilities}} \\ &= \frac{60\,000}{20\,000} = \frac{3}{1} = \mathbf{3:1} \end{aligned}$$

The current ratio of **3:1** means that the creditors are secured since for every ₱ 1.00 of debt, the company has ₱ 3.00 available for payment.

$$\begin{aligned} \text{b. Debt Ratio} &= \frac{\text{Total Liabilities}}{\text{Total Assets}} \\ &= \frac{350\,000}{700\,000} = \frac{1}{2} = \mathbf{1:2} \end{aligned}$$

The debt ratio of **1:2** means that the company is secured since it can pay its obligations. Further, the company has ₱ 1.00 liability for ₱ 2.00 of total assets.



NOTE

Ratio is a comparison of two or more positive quantities.

a and ***b*** are two quantities and ***b* ≠ 0**, the ratio of ***a*** to ***b*** is written as

$$\mathbf{a:b} \text{ or } \frac{\mathbf{a}}{\mathbf{b}}.$$

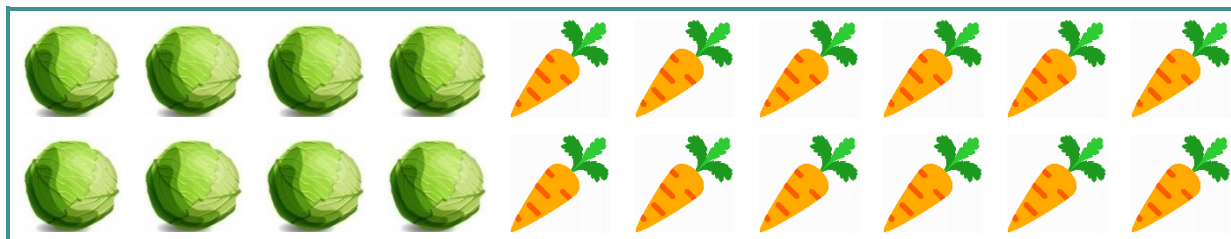
- ***a*** → first quantity of a ratio
 - ***b*** → second quantity of a ratio
1. We read the ratio **2 : 3** as **2 to 3**.
 2. Ratio does not have a unit.
 3. A colon (**:**) is used to express ratio.
 4. Changing the order of quantities in a ratio gives rise to a different ratio.

Lesson 1.2: Equivalent Ratios

The two or three quantities in a ratio share a multiplicative relationship. This means that when one quantity is changed, the other quantities will change proportionately.

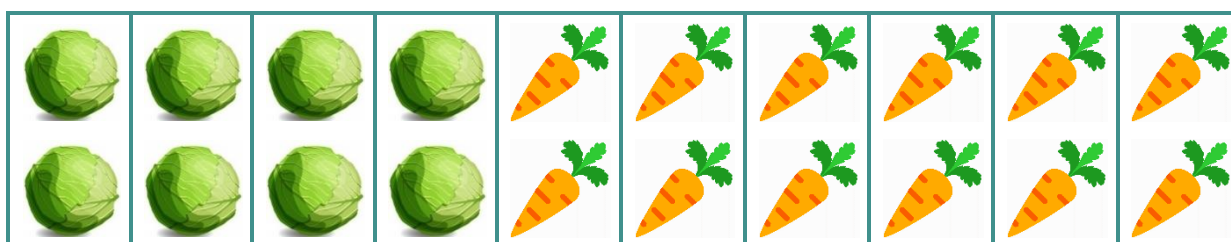
Illustrative example:

Ivan sold 8 heads of cabbage and 12 pieces of carrots harvested from his farm.



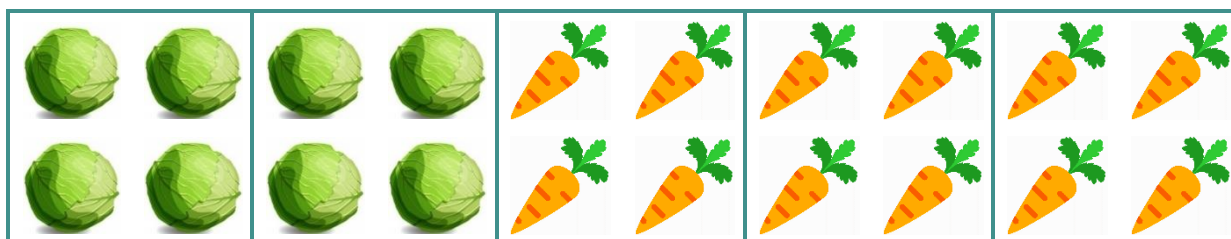
The ratio of the number of cabbages to the number of carrots sold is 8:12.

Ivan packed the sold vegetables in groups of two.



The ratio of the number of packs of cabbages to the packs of carrots is 4:6.

A moment after, Ivan decided to repack the sold vegetables in groups of 4.



The ratio of the number of packs of cabbages to the packs of carrots is 2:3.



NOTE

- $8 : 12$, $4 : 6$, and $2 : 3$ are **equivalent ratios**, thus,
 $8 : 12 = 4 : 6 = 2 : 3$
- $2 : 3$ is a ratio in its **simplest form**.
- In expressing ratios in their simplest form, simply divide the terms by their greatest common factor.
- A ratio remains unchanged if we multiply or divide its terms by the same number.

LESSON 2: PROPORTION

Lesson 2.1: Introduction to Proportion

Having an idea on what ratio is will help you have a better understanding on what proportion is and its different kinds.

Study the example below:

Suppose a dozen of mangoes costs ₱ 150 and you have sold 3 dozens of mangoes. How much is your sales?

Dozen of mangoes : cost of one dozen = 3 dozens of mangoes: cost of 3 dozens

$$\frac{1 \text{ dozen of mangoes}}{\text{₱ 150 per dozen}} = \frac{3 \text{ dozens of mangoes}}{x} \Rightarrow 1x = 3(150) \Rightarrow x = 450$$

Therefore, the cost of 3 dozens of mangoes is ₱ 450.



NOTE

Proportion refers to the equality between ratios. Which can be written in two ways:

- two equal fractions

$$\frac{a}{b} = \frac{c}{d} \quad \text{or} \quad \frac{3}{4} = \frac{6}{8}$$

- using a colon

$$a : b = c : d \quad \text{or} \quad 3 : 4 = 6 : 8$$

means
extremes

If the product of the means is equal to the product of the extremes, then we have a proportion.

$$a \times d = b \times c \quad \text{or} \quad 3(8) = 4(6) \Rightarrow 24 = 24$$

Proportions are of three types:

1. **Direct proportion** – a number is directly proportionate to another when as one value increases, so does the other.
2. **Indirect/inverse proportion** – a number is indirectly proportionate to another when as one value increases, the other decreases.
3. **Partitive proportion** – involves identifying parts of a whole based on a given ratio of these parts.

Lesson 2.2: Kinds of Proportion

A. DIRECT PROPORTION

Two values x and y are directly proportional to each other when x and y will either increase or decrease together. Meaning, when one increases, the other also increases or when one decreases, the other decreases as well.

Here are real-life situations of direct proportion:

- Time worked and salary at an hourly rate
 - the longer the time you work, the higher salary you will be paid.
- Monthly salary and the maximum amount you can borrow from a lending institution.
 - a higher monthly salary allows you to loan or borrow a higher amount of money.

B. INDIRECT or INVERSE PROPORTION

Two values x and y are inversely proportional to each other such that if x increases, y decreases or if x decreases, y increases.

Here are real-life situations of indirect proportion:

- Speed and time
 - a faster speed can cover a certain distance in a shorter period of time, while a slower speed results to a longer period of time.
- Expenses and net income
 - a high amount of expenses in a certain business yields to a lower net income.

C. PARTITIVE PROPORTION

In partitive proportion, we are talking about a whole divided into parts or a whole being partitioned into equal or unequal ratios.

Here are real-life situations of partitive proportion:

- For a certain recipe, Chris combines water and vinegar at a ratio 3 : 11.
- A father wants to leave ₱ 2 500 000 to his four children in the ratio 2 : 3 : 3 : 2.
- A profit of ₱ 100 000 is to be divided among three business partners according to their capital ratio which is 5: 3: 2.



What's More



Activity #2: More of Ratio and Proportion

A. Write ratios in colon (:) form based on the given situation.

A basketball team won 12 games and lost 3 games.

1. What is the ratio of games won to total of games played?
2. What is the ratio of games won to the games lost?
3. What is the ratio of games lost to games played?
4. What is the ratio of games lost to games won?

B. Write the appropriate proportion statement in colon form (:) for the following problems by letting x be the unknown, but do not solve for x .

1. Three pieces of black marker cost ₱ 32. How much do 10 pieces of black marker cost?
2. The ratio of the number of pens to the number of sheets of paper is 4 : 8. If there are 40 sheets of paper, how many pens are there?
3. Anna can type 140 words in 3 minutes. How many words can she type in 10 minutes?
4. The ratio of boys to girls in a badminton game is 4 : 3. Marie counted that there are 12 more boys than girls. How many boys and girls are there in the tournament?
5. Three men can finish doing the interior designing of a house in 3 weeks. How many men is needed to finish the interior designing in a week?



Activity #3: Proportionality in Everyday Life

Direction: List two (2) real-life situations that exemplifies direct, inverse, and partitive proportion. Use a separate sheet for your answers. An example is given for each.

A. Direct Proportion

Example:

Monthly average expenditure and number of family members – a small family spends less, while a big family spends more.

1. _____

2. _____

B. Indirect or Inverse Proportion

Example:

Price and quantity – given a specific budget, a cheaper price allows you to purchase a greater quantity of an item compared to an expensive one.

1. _____

2. _____

C. Indirect or Inverse Proportion

Example:

Three businessmen share profits and losses at a ratio of 2 : 3 : 5.

1. _____

2. _____

~ DAY 4 ~



What I Have Learned



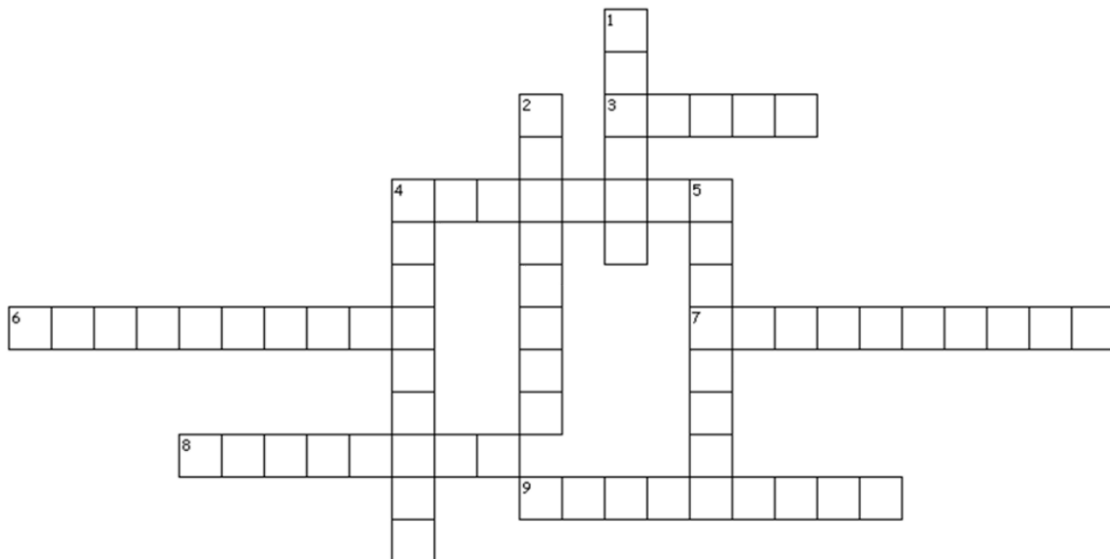
NOTE

A **ratio** is a way to compare quantities while **proportion** on the other hand is an equation that says that two ratios are equal. Thus, if one quantity is unknown in a proportion, you can find it by solving the proportion. Considering the type of proportion illustrated in a given situation will further help you find the missing quantity.



Activity #4: Let's Sum It Up!

Direction: Complete the puzzle below by using the clues given.



Across

3. It is the comparison of two or more positive quantities.
4. A proportion is established when the _____ of means and extremes are equal.
6. 5:2 and 20:8 are said to be _____ ratios.
7. It refers to the equality between ratios.
8. Multiplying or _____ the terms of a ratio with the same number does not change the ratio.
9. Aside from using colon to write a proportion, we can also use two equal _____.

Down

1. A proportion shown if the increase in customer results to an increase in earning of a restaurant.
2. A type of proportion exemplified when the duration to complete a particular project is shortened due to additional workers.
4. A proportion which identifies a portion of a whole based on given ratios.
5. Writing 6:42 as 1:7 is in its _____ form.



Being able to see the relevance of our lesson into real-life leads us to better understanding and appreciation of the concepts of ratio and proportion. Here are real-life problems where ratio and proportion show their importance.

Carefully read each problem, analyze, and apply your knowledge on ratio and proportion in answering the questions in each given situation. Use a separate sheet for your answers.

Problem #1:

Darwin and Erik are business partners. Their invested capital is ₱ 30 000 and ₱ 45 000, respectively. What is their capital ratio?

Problem #2:

Current ratio shows if a company has enough current assets to meet its current liabilities ($\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$) while debt ratio shows the relationship between the total liabilities and the total assets ($\text{Debt Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$).

Assuming a firm has ₱ 900 000.00 current assets, ₱ 1 200 000.00 total assets, ₱ 300 000.00 current liabilities, and ₱ 600 000.00 total liabilities.

- Find the current ratio and debt ratio.
- What do the current ratio and debt ratio imply? (Interpretation of the ratios in a.)

Post-Assessment



Multiple Choice: Choose the correct answer to each question by writing the letter of your choice. Remember to write your answers on a separate sheet of paper.

1. Mario had 28 bottles of strawberry jam for sale. He sold 7 bottles of the strawberry jam to a friend. What is the ratio of the number of bottles of strawberry jam sold to that of the number of bottles of strawberry jam left?
A. 1:3 B. 1:4 C. 7:21 D. 7:28
2. Express the ratio 324:252:180 in simplest form.
A. 162:126:90 C. 27:21:15
B. 81:63:45 D. 9:7:5
3. Which of the following ratios are equivalent?
A. 300 miles in 6 hours and 200 miles in 5 hours.
B. 8 km in 21 minutes and 12 km in 31.5 minutes.
C. 5 adults to accompany 25 children and 3 adults to accompany 12 children.
D. 5 glasses of iced tea for 180 grams of powdered mix and 2 glasses of iced tea for 36 grams of powdered mix.
4. What value of x will make the fractions $\frac{9}{21}$ and $\frac{x}{231}$ equivalent?
A. 90 B. 99 C. 108 D. 117
5. How many meters of ribbon will be needed for a dozen gifts, if a gift box needs $\frac{19}{24}$ m of ribbon?
A. 9 m B. 9.5 m C. 18 m D. 19 m
6. Which of the following statements is true about the kinds of proportion?
A. A number is directly proportionate to another when as one value increases, the other decreases.
B. A number is indirectly proportionate to another when as one value increases, so does the other.
C. Partitive proportion involves identifying parts of a whole based on a given ratio of these parts.
D. Direct proportion has two values that may vary so the ratio between them also varies.
7. A type of proportion in which the quantities are inversely related with each other.
A. Direct Proportion C. Mere Proportion
B. Indirect Proportion D. Partitive Proportion

8. Which of the given quantities show direct relationship?
- | | |
|--------------------------------|-------------------------------|
| A. Expenses and Income | D. Number of workers and Days |
| B. Volume and Pressure | |
| C. Price and Quantity Supplied | |
9. What kind of proportion is exemplified in the situation: "Three businessmen share profits and losses at a ratio of 2:3:5."?
- | | |
|-------------|--------------|
| A. Direct | C. Mere |
| B. Indirect | D. Partitive |
10. Which of the following real-life examples is true about proportion?
- Time and speed are directly related.
 - Quantity and cost are indirectly related
 - Number of workers and days is under partitive proportion.
 - Budgeting a monthly salary is an example of partitive proportion.
11. Kobe resells facemasks and face shields in this time of pandemic in the ratio 8:3. If he sold 121 facemasks and face shields for the past week, how many facemasks did he sell?
- | | | | |
|-------|-------|-------|-------|
| A. 22 | B. 33 | C. 66 | D. 88 |
|-------|-------|-------|-------|
12. The quality control team of a toy company accounted that there are 5 defective toy models for every 500 models. If the company makes 66 000 toy models annually, how many of them are expected to be defective?
- | | | | |
|--------|--------|--------|--------|
| A. 330 | B. 440 | C. 550 | D. 660 |
|--------|--------|--------|--------|
13. Sam bought 15 cones of ice cream with 3 scoops of different flavors in each cone. How many scoops of ice cream will there be if there are 150 cones?
- | | | | |
|-------|--------|--------|----------|
| A. 30 | B. 450 | C. 750 | D. 2 250 |
|-------|--------|--------|----------|
14. Ciara and her friends started to save money to be donated to the front liners in their province. In two weeks, they were able to save ₱ 6 000. With this rate of saving money, how many weeks will they need to be able to raise ₱ 78 000?
- | | |
|-------------|-------------|
| A. 26 weeks | C. 15 weeks |
| B. 24 weeks | D. 13 weeks |
15. Justine and Ethan agreed to go into a partnership. The former invested ₱ 60 000 while the latter invested ₱ 120 000. What part of the total capital is that of Justine?
- | | | | |
|------------------|------------------|------------------|------------------|
| A. $\frac{2}{3}$ | B. $\frac{1}{2}$ | C. $\frac{1}{4}$ | D. $\frac{1}{3}$ |
|------------------|------------------|------------------|------------------|

~ DAY 5 ~

Additional Activities



Activity #5: Let's Make It Simple

Direction: Copy the given ratios and simplify. Also, eliminate decimals and fractions in your final answers.

1. 5 : 25
2. 72 : 64
3. 3 : 12 : 24
4. 81 : 27 : 45
5. 9 : 12 : 36
6. 5 pesos : 50 cents
7. 1 200 minutes : 20 seconds
8. $2\frac{2}{3}$ minutes : $15\frac{1}{4}$ hours
9. 0.36 km : 680 cm
10. 3km^2 : $21\,000\text{m}^2$



Activity #6: Are They Equivalent Ratios?

Direction: Read and follow the instruction in each part of this activity. Use a separate sheet for your answers.

- A. Determine whether the following ratios are equivalent or not.
Write = if they are equivalent and $\neq$ if they are not.

- _____ 1. 300 miles in 6 hours and 250 miles in 5 hours
- _____ 2. 5 adults to look after 20 children and 3 adults to look after 12 children
- _____ 3. 8 meters in 21 minutes and 12 meters in 31.5 minutes
- _____ 4. ₱ 500 for 5 pairs of socks and ₱ 300 for 2 pairs of socks
- _____ 5. 5 glasses of iced tea for 180 grams of powdered mix and 2 glasses of iced tea for 36 grams of powdered mix.

- B. Find the value of x so that the following fractions are equivalent.

1. $\frac{5}{6}$ and $\frac{x}{18}$
2. $\frac{15}{x}$ and $\frac{45}{75}$

3. $\frac{288}{336}$ and $\frac{24}{x}$

5. $\frac{175}{500}$ and $\frac{x}{50}$

4. $\frac{x}{21}$ and $\frac{171}{399}$

C. Find the missing numbers.

1. $8 : 3 : 5 = _ : 9 : 15$

2. $16 : 28 : 4 = 8 : 14 : _$

3. $14 : 7 : 35 = 2 : _ : 5$

4. $81 : 63 : 45 = _ : _ : 5$

5. $24 : 42 : 54 = _ : _ : _$ (*answer in lowest terms*)



Activity #7: Slight Problems

Direction: Read and understand the following problems. Use a separate sheet to organize your solution.

1. The birth rate in a certain barangay is 25 births per 1 000 population per year. How many births are there in one year if this birth rate is also true for the whole country whose population is now 104 million?

2. Andy and Bill put up a business with a starting capital shares of ₱ 60 000 and ₱ 90 000 respectively.

a. What is their capital ratio?

b. If they agreed to share profits and losses in accordance with their capital ratio, how much is the total profit for the first month of operation if Bill receives ₱ 30 000?

Answer Key



What I Know 1. A 2. C 3. B 4. B 5. A 6. C 7. B 8. D 9. D 10. D 11. B 12. C 13. C 14. A 15. A What's New Activity #1: Help Me Write Ratios Answers may vary. What's More Activity 2: More of Ratio and Proportion A. 1. 4:5 2. 4:1 3. 1:5 4. 1:4 5. 3:x=1:3 Activity #3: Proportionality in Everyday Life Answers may vary.	What I Have Learned Activity #4: Let's Sum It Up! Across Down 3. Ratio 4. Products 6. Equivalent 7. Proportion 8. Dividing 9. Fractions What I Can Do Problem #1 2:3 Problem #2 a. 3:1 and 1:2 b. Current Ratio: The firm is capable in paying its debts; thus, the creditors are secured. The firm has available ₱ 3.00 for every ₱ 1.00 of its debt. Debt Ratio: The firm has ₱ 1.00 liability for ₱ 2.00 total assets, meaning, the firm is secured since it can pay its obligations. Post-Assessment 1. C 2. D 3. B 4. B 5. B 6. C 7. B 8. C 9. D 10. D 11. D 12. D 13. B 14. A 15. D	Additional Activities Activity #5: Let's Make It Simple 1. 1:5 2. 9:8 3. 1:4:8 4. 9:3:5 5. 3:4:12 6. 10:1 7. 3 600:1 8. 32:183 9. 900:17 10. 1 000:7 Activity #6: Are They Equivalent Ratios A. 1. = 2. = 3. = 4. ≠ 5. ≠ B. 1. 15 2. 25 3. 28 4. 9 5. 17.5 C. 1. 24 2. 2 3. 1 4. 9 : 7 5. 4 : 7 ; 9 Activity #7: A Slight Problem 1. 2 600 000 2. a. 2:3 b. ₱ 50 000
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